



Faculty of Engineering and Technology
Department of Electrical and Computer Engineering

ENCS3330
Digital Integrated Circuits

Assignment 2

Prepared by: Karim Taha
Student Number: 1211155
Instructor: Dr. Alhareth Zyoud
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Abstract

A CMOS pass gate, a tri-state inverter, a P-latch, an N-latch, and a master–slave D Flip-Flop (P-latch + N-latch) are designed and simulated in LTspice. Each block is simulated at the schematic and layout level, and rise/fall times are compared.

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1 Pass Gate

1.1 Schematic Design and Simulation

The pass gate is a PMOS and an NMOS connected in parallel between input A and output OUT , controlled by complementary signals C and $\overline{C}$.

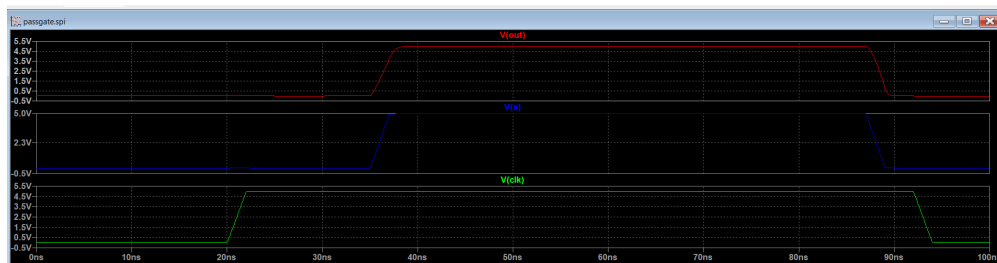
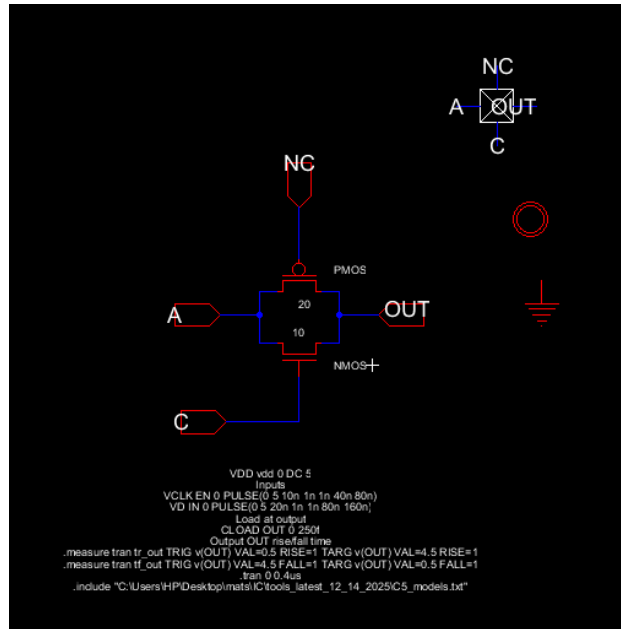


Figure 1: Pass gate schematic (top) and its simulation waveforms (bottom).

1.2 Layout Design and Simulation

The PMOS and NMOS are placed side by side, sharing the input/output diffusion contacts.

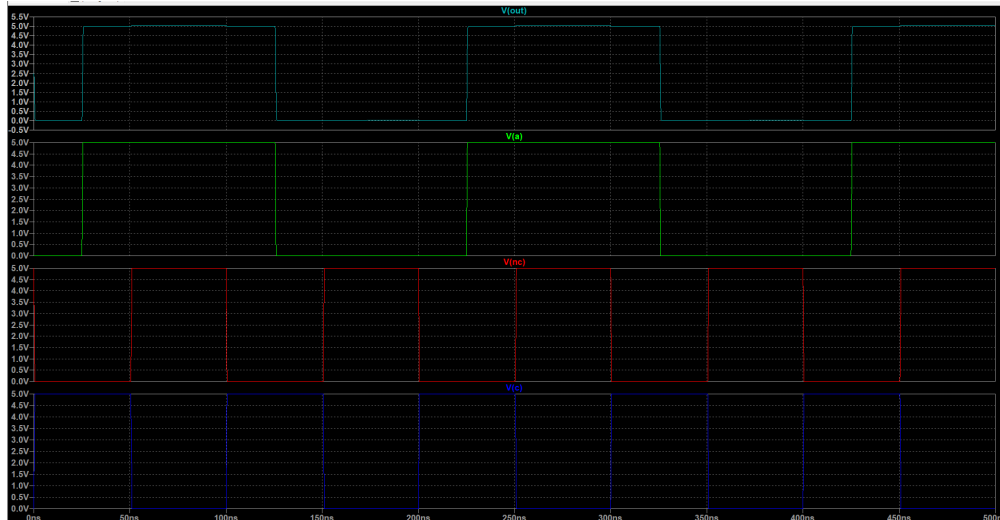
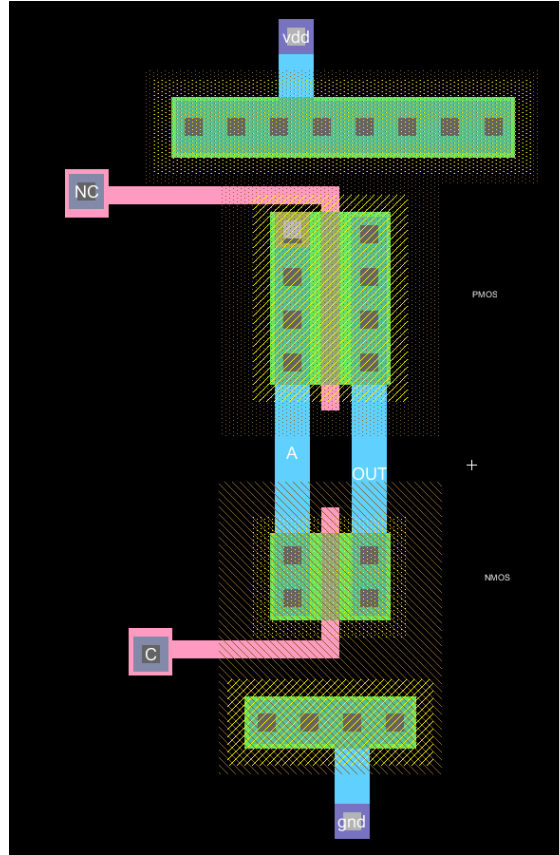


Figure 2: Pass gate layout (top) and its simulation waveforms (bottom).

2 Tri-State Inverter

2.1 Schematic Design and Simulation

The tri-state inverter is a stack of two PMOS and two NMOS transistors, with the inner pair driven by IN and the outer pair driven by $EN/\overline{EN}$. When disabled, the output is high impedance.

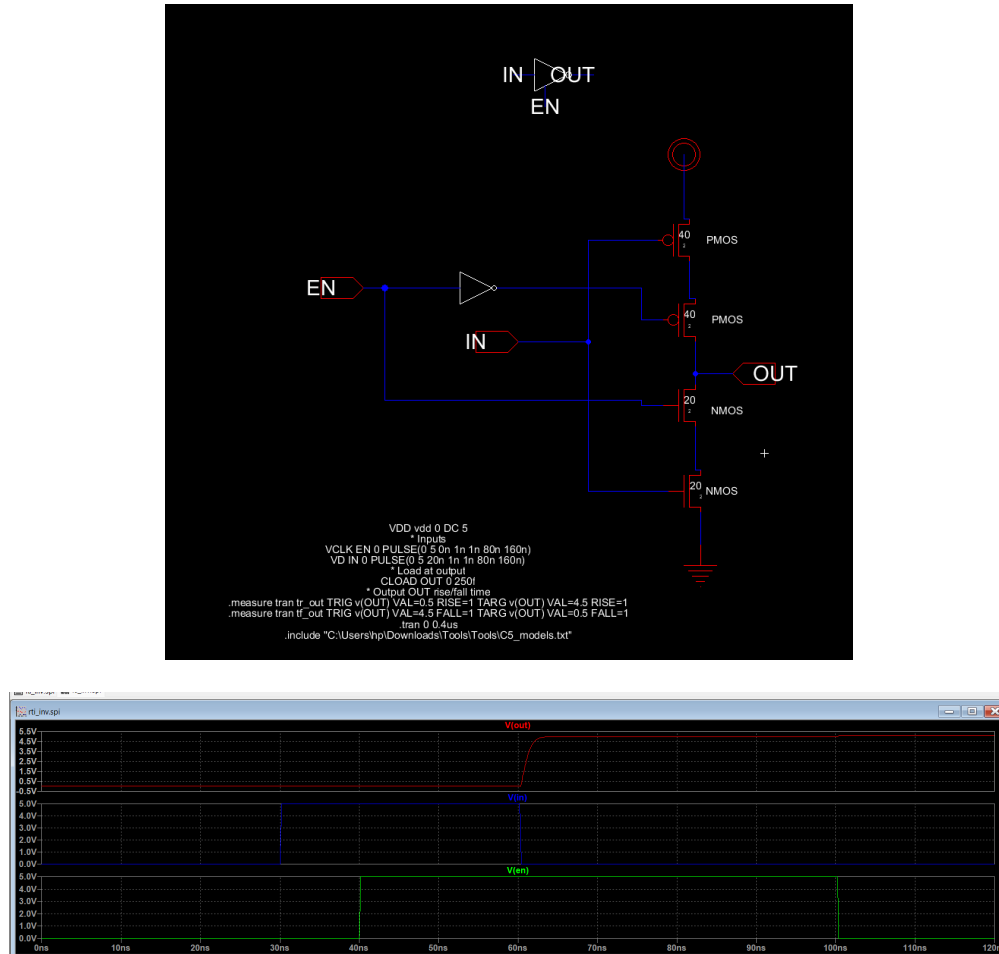


Figure 3: Tri-state inverter schematic (top) and its simulation waveforms (bottom).

2.2 Layout Design and Simulation

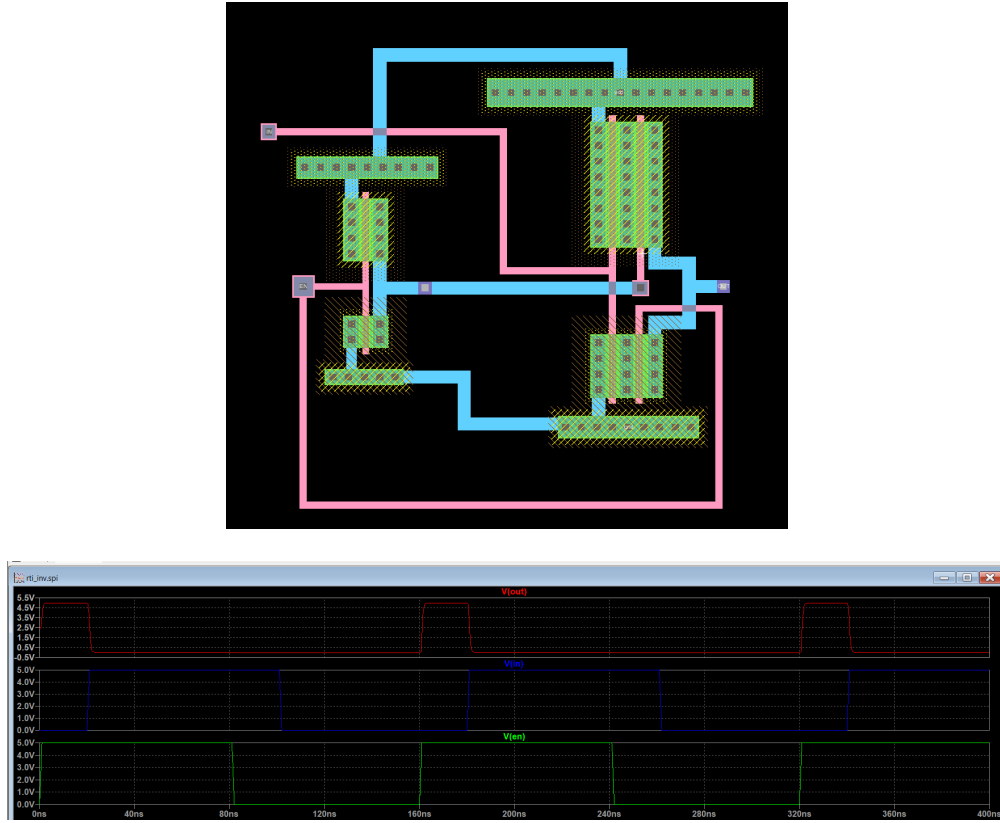


Figure 4: Tri-state inverter layout (top) and its simulation waveforms (bottom).

3 P-Latch

3.1 Schematic Design and Simulation

The P-latch is a pass gate followed by two cross-coupled inverters (storage loop) and an output buffer, clocked directly by $CLK/\overline{CLK}$.

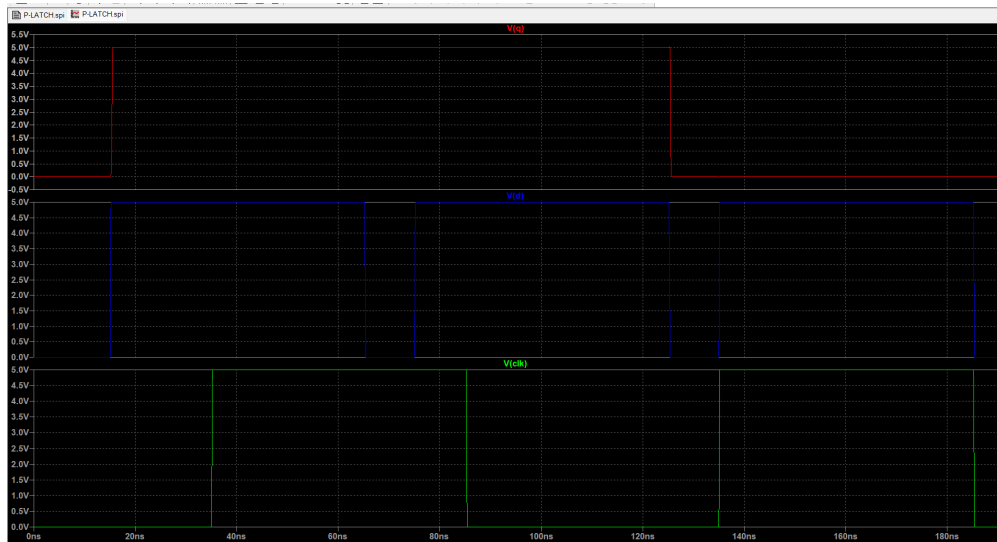
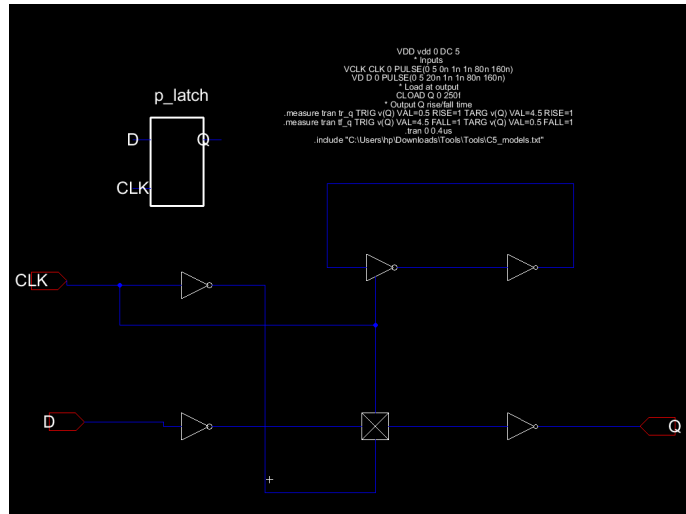


Figure 5: P-latch schematic (top) and its simulation waveforms (bottom).

3.2 Layout Design and Simulation

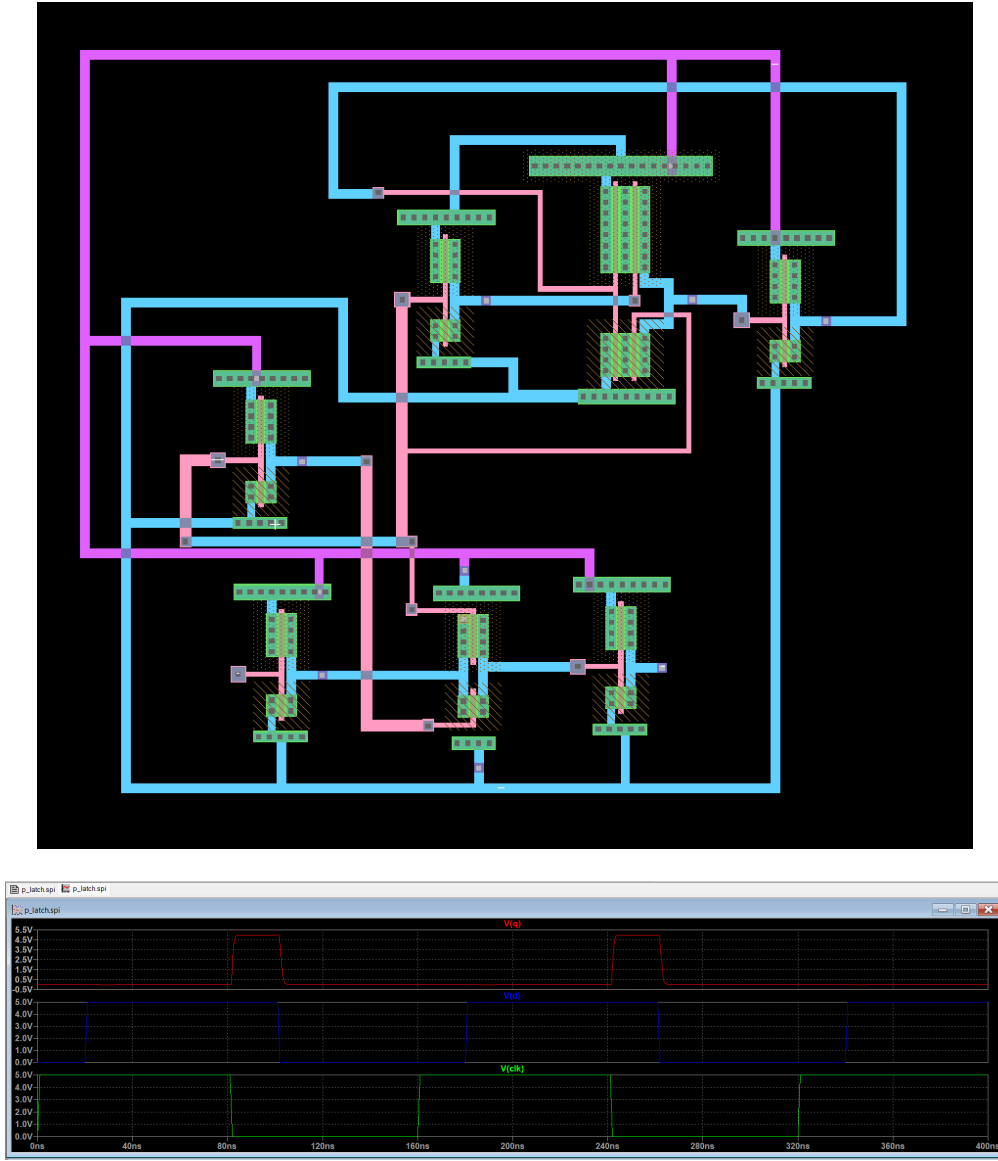


Figure 6: P-latch layout (top) and its simulation waveforms (bottom).

4 N-Latch

4.1 Schematic Design and Simulation

The N-latch has the same structure as the P-latch, but its pass gate and feedback inverter are clocked with the opposite phase, so it samples while the P-latch holds, and vice versa.

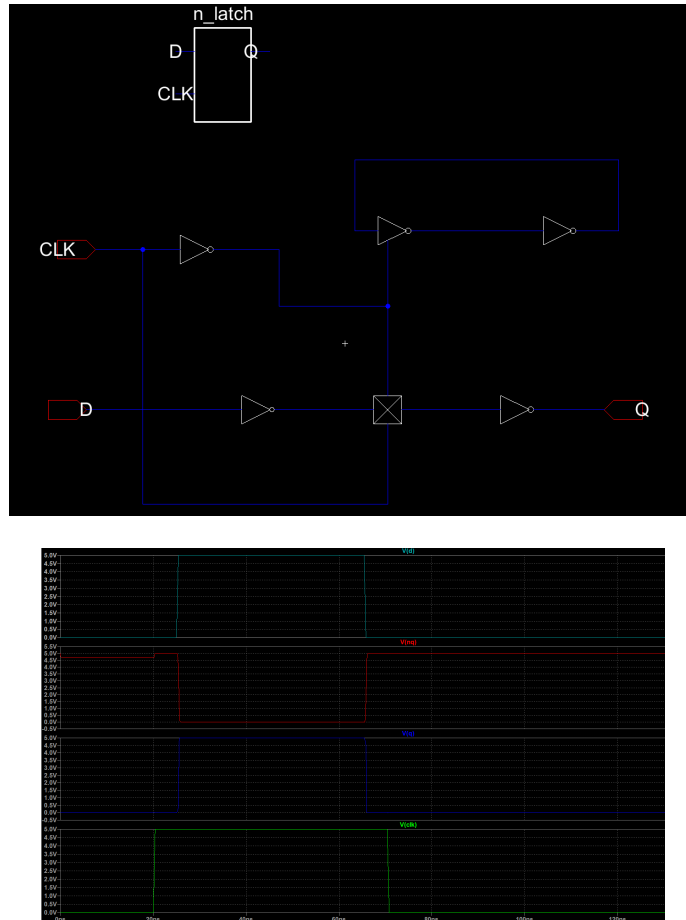


Figure 7: N-latch schematic (top) and its simulation waveforms (bottom).

4.2 Layout Design and Simulation

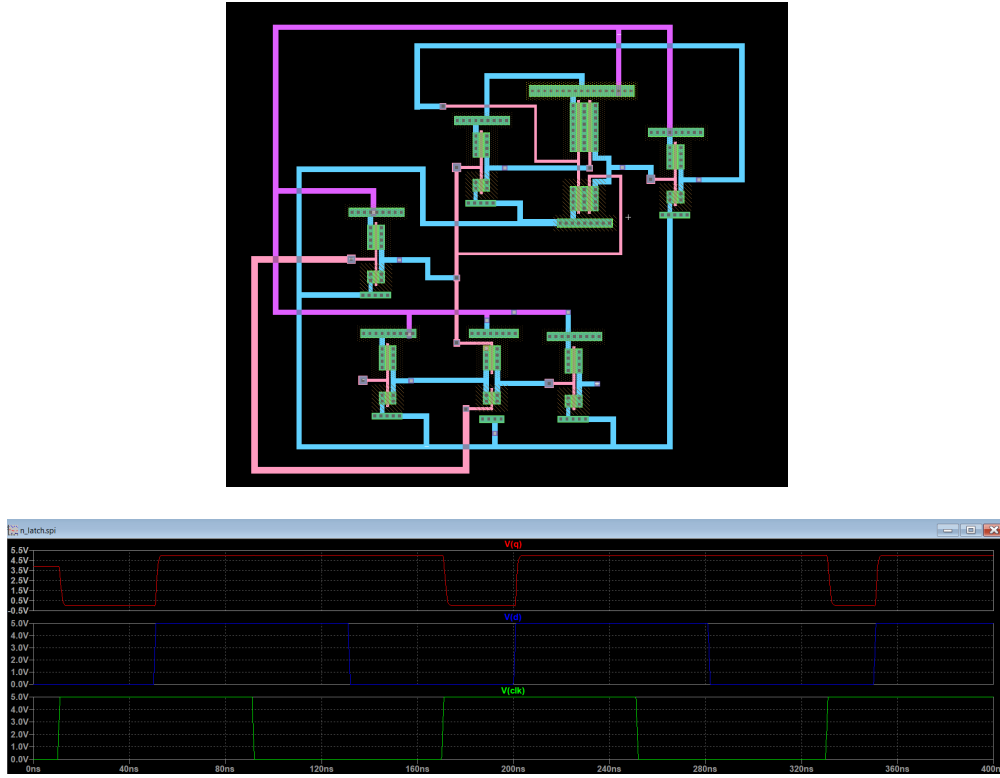


Figure 8: N-latch layout (top) and its simulation waveforms (bottom).

5 D Flip-Flop (P-Latch + N-Latch)

5.1 Schematic Design and Simulation

The DFF cascades the P-latch (master) and the N-latch (slave) on the same clock. The master samples D while the slave holds, then the slave samples the master's output while the master holds, giving one edge-triggered update of Q per clock cycle.

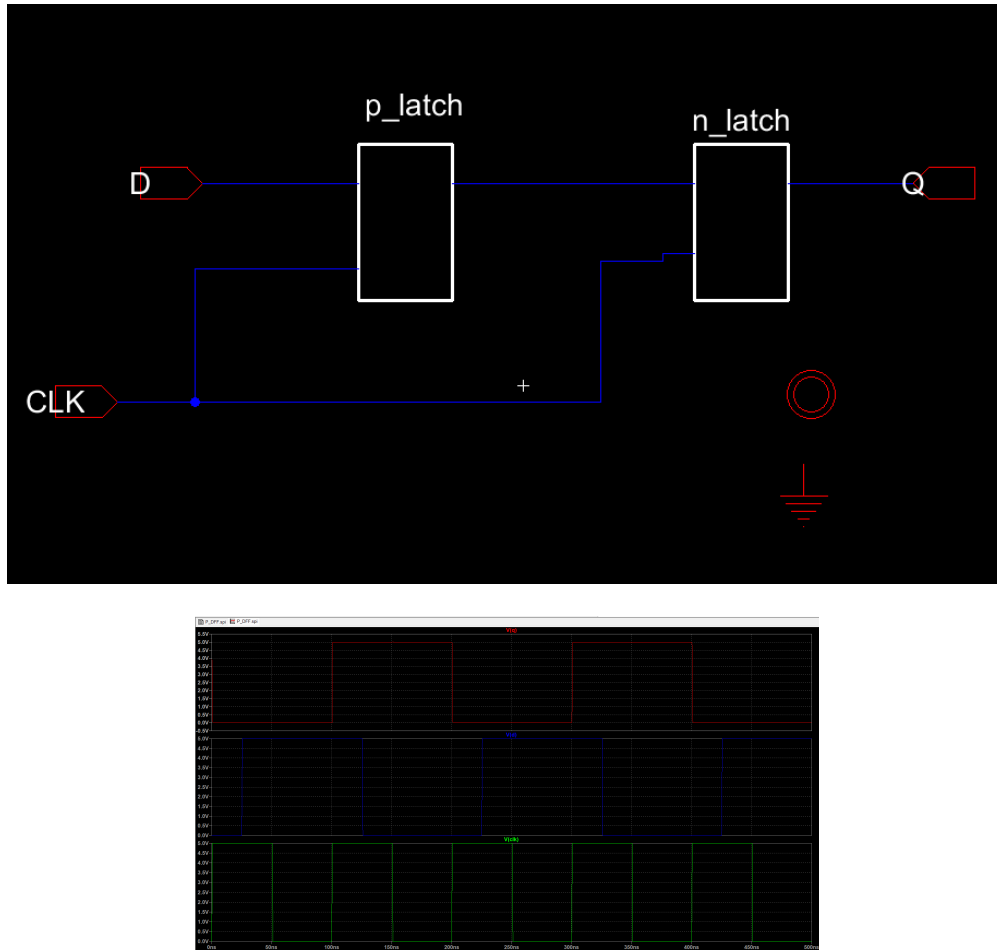


Figure 9: D Flip-Flop schematic – P-latch (master) + N-latch (slave) – (top) and its simulation waveforms (bottom).

5.2 Layout Design and Simulation

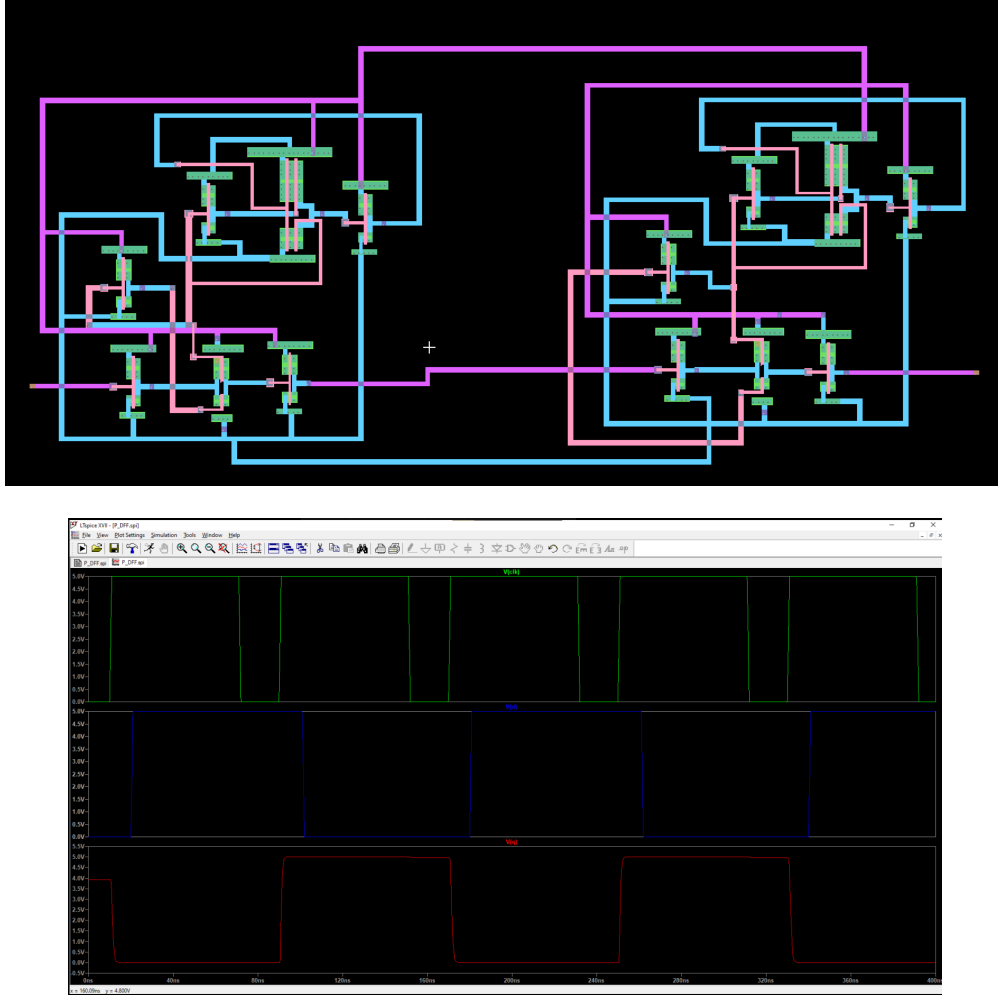
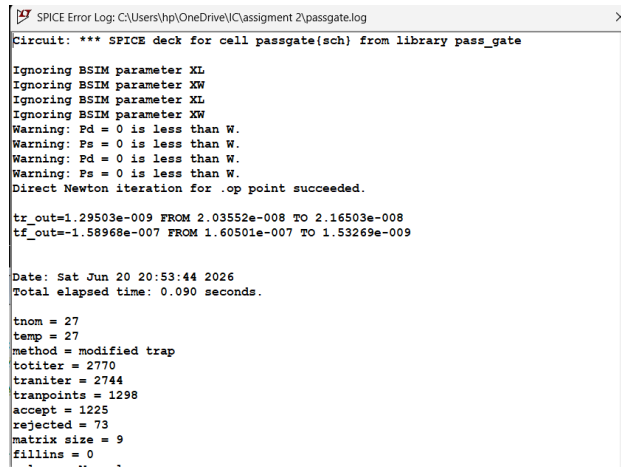


Figure 10: D Flip-Flop layout (top) and its simulation waveforms (bottom).

6 Timing Results and Comparison

Rise time (t_r) and fall time (t_f) are measured between the 10% and 90% crossings of the output. Both schematic and post-layout extracted simulations are compared below.

6.1 Pass Gate

A screenshot of a SPICE Error Log window titled "SPICE Error Log: C:\Users\hnp\OneDrive\Class\assignment 2\passgate.log". The log contains the following text:

```
Circuit: *** SPICE deck for cell passgate(sch) from library pass_gate

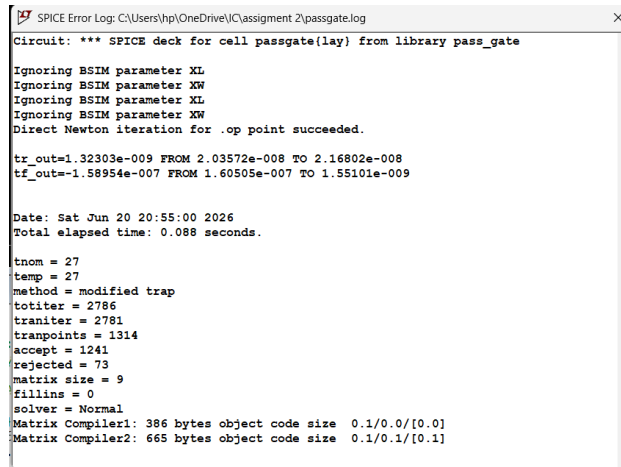
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Direct Newton iteration for .op point succeeded.

tr_out=1.29503e-009 FROM 2.03552e-008 TO 2.16503e-008
tf_out=-1.58968e-007 FROM 1.60501e-007 TO 1.53269e-009

Date: Sat Jun 20 20:53:44 2026
Total elapsed time: 0.090 seconds.

tnom = 27
temp = 27
method = modified trap
totiter = 2770
traniter = 2744
tranpoints = 1298
accept = 1225
rejected = 73
matrix size = 9
fillins = 0
```

Figure 11: Pass gate schematic – rise/fall measurement.

A screenshot of a SPICE Error Log window titled "SPICE Error Log: C:\Users\hnp\OneDrive\Class\assignment 2\passgate.log". The log contains the following text:

```
Circuit: *** SPICE deck for cell passgate(layout) from library pass_gate

Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Direct Newton iteration for .op point succeeded.

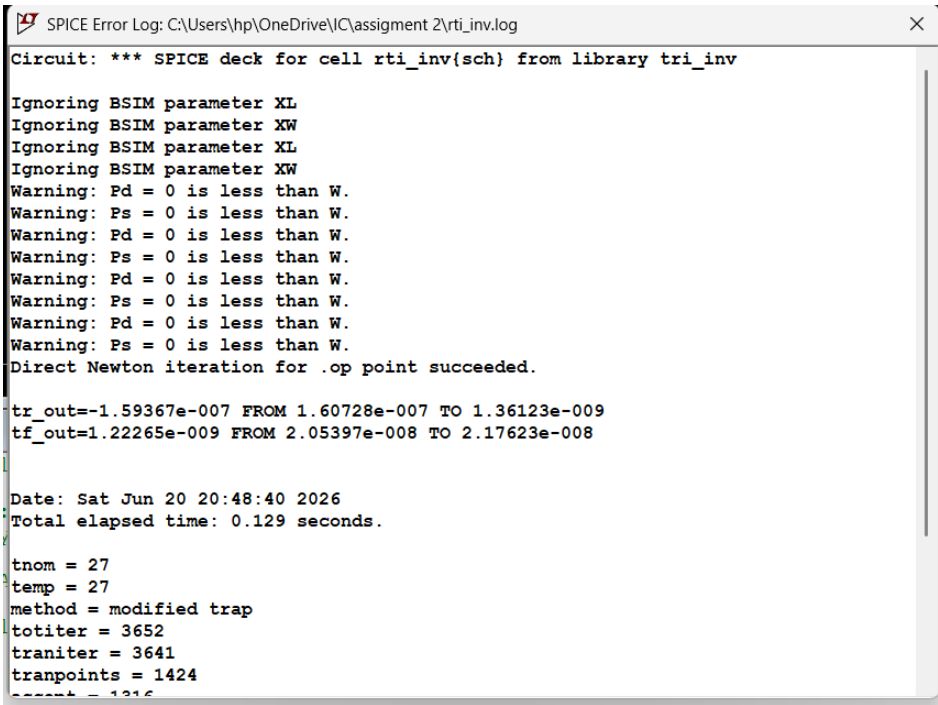
tr_out=1.32303e-009 FROM 2.03572e-008 TO 2.16802e-008
tf_out=-1.58954e-007 FROM 1.60505e-007 TO 1.55101e-009

Date: Sat Jun 20 20:55:00 2026
Total elapsed time: 0.088 seconds.

tnom = 27
temp = 27
method = modified trap
totiter = 2786
traniter = 2781
tranpoints = 1314
accept = 1241
rejected = 73
matrix size = 9
fillins = 0
solver = Normal
Matrix Compiler1: 386 bytes object code size 0.1/0.0/[0.0]
Matrix Compiler2: 665 bytes object code size 0.1/0.1/[0.1]
```

Figure 12: Pass gate layout – rise/fall measurement.

6.2 Tri-State Inverter

A screenshot of a SPICE Error Log window titled "SPICE Error Log: C:\Users\hp\OneDrive\IC\assignment 2\rti_inv.log". The log contains the following text:

```
Circuit: *** SPICE deck for cell rti_inv{sch} from library tri_inv

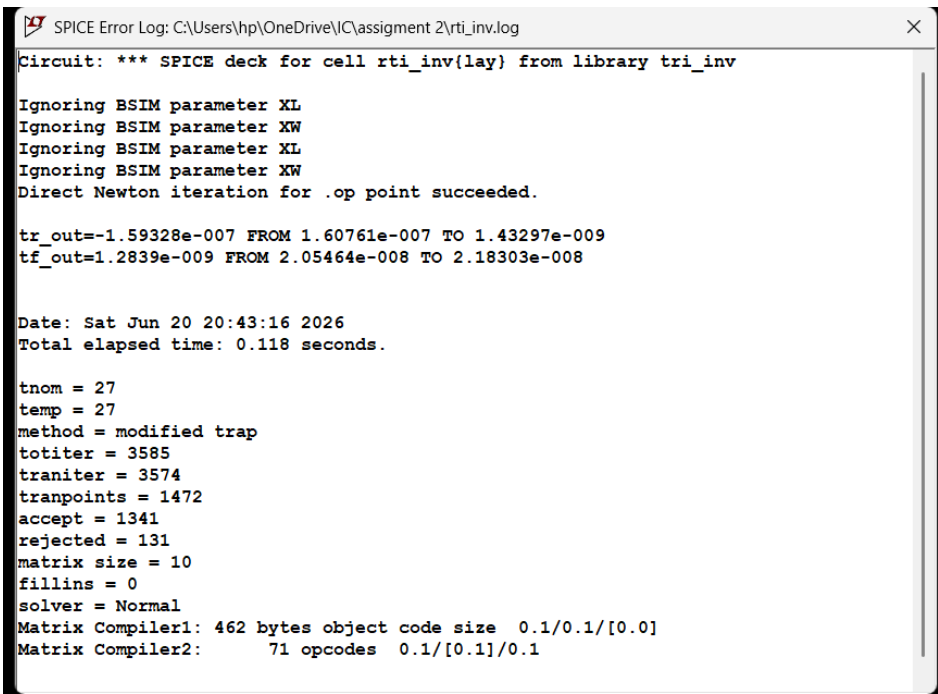
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Direct Newton iteration for .op point succeeded.

tr_out=-1.59367e-007 FROM 1.60728e-007 TO 1.36123e-009
tf_out=1.22265e-009 FROM 2.05397e-008 TO 2.17623e-008

Date: Sat Jun 20 20:48:40 2026
Total elapsed time: 0.129 seconds.

tnom = 27
temp = 27
method = modified trap
totiter = 3652
traniter = 3641
tranpoints = 1424
accept = 1316
```

Figure 13: Tri-state inverter schematic – rise/fall measurement.

A screenshot of a SPICE Error Log window titled "SPICE Error Log: C:\Users\hp\OneDrive\IC\assignment 2\rti_inv.log". The log contains the following text:

```
Circuit: *** SPICE deck for cell rti_inv{lay} from library tri_inv

Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Direct Newton iteration for .op point succeeded.

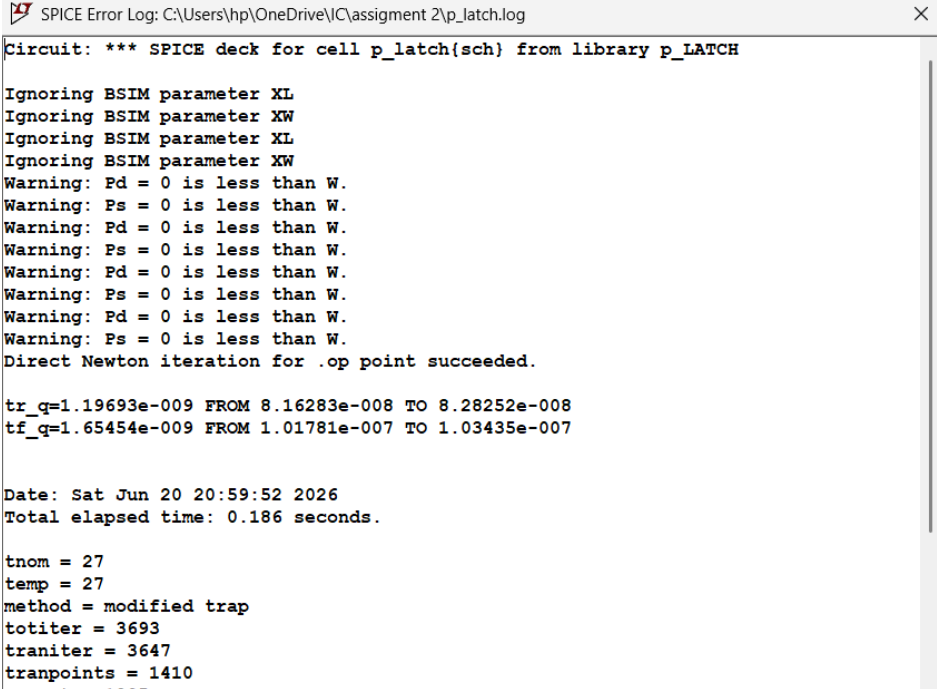
tr_out=-1.59328e-007 FROM 1.60761e-007 TO 1.43297e-009
tf_out=1.2839e-009 FROM 2.05464e-008 TO 2.18303e-008

Date: Sat Jun 20 20:43:16 2026
Total elapsed time: 0.118 seconds.

tnom = 27
temp = 27
method = modified trap
totiter = 3585
traniter = 3574
tranpoints = 1472
accept = 1341
rejected = 131
matrix size = 10
fillins = 0
solver = Normal
Matrix Compiler1: 462 bytes object code size 0.1/0.1/[0.0]
Matrix Compiler2: 71 opcodes 0.1/[0.1]/0.1
```

Figure 14: Tri-state inverter layout – rise/fall measurement.

6.3 P-Latch



```
SPICE Error Log: C:\Users\hp\OneDrive\IC\assignment 2\p_latch.log
Circuit: *** SPICE deck for cell p_latch{sch} from library p_LATCH

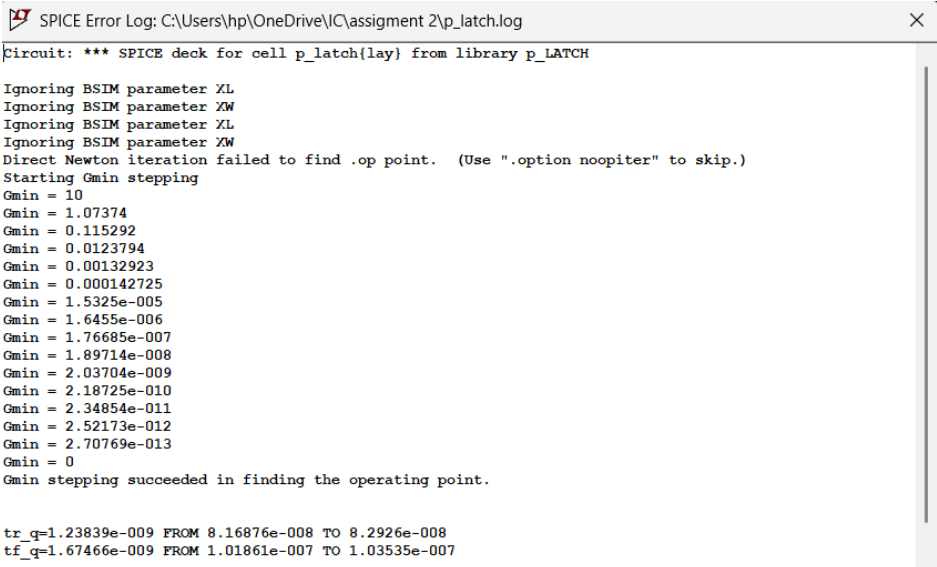
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Direct Newton iteration for .op point succeeded.

tr_q=1.19693e-009 FROM 8.16283e-008 TO 8.28252e-008
tf_q=1.65454e-009 FROM 1.01781e-007 TO 1.03435e-007

Date: Sat Jun 20 20:59:52 2026
Total elapsed time: 0.186 seconds.

tnom = 27
temp = 27
method = modified trap
totiter = 3693
traniter = 3647
tranpoints = 1410
```

Figure 15: P-latch schematic – rise/fall measurement.



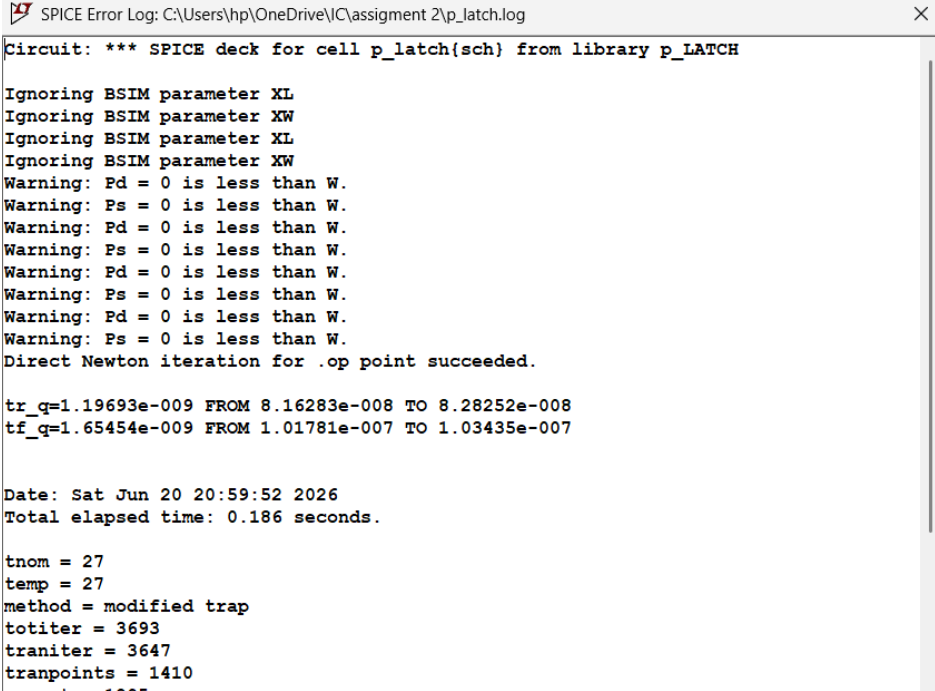
```
SPICE Error Log: C:\Users\hp\OneDrive\IC\assignment 2\p_latch.log
Circuit: *** SPICE deck for cell p_latch{lay} from library p_LATCH

Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Direct Newton iteration failed to find .op point. (Use ".option noopiter" to skip.)
Starting Gmin stepping
Gmin = 10
Gmin = 1.07374
Gmin = 0.115292
Gmin = 0.0123794
Gmin = 0.00132923
Gmin = 0.000142725
Gmin = 1.5325e-005
Gmin = 1.6455e-006
Gmin = 1.76685e-007
Gmin = 1.89714e-008
Gmin = 2.03704e-009
Gmin = 2.18725e-010
Gmin = 2.34854e-011
Gmin = 2.52173e-012
Gmin = 2.70769e-013
Gmin = 0
Gmin stepping succeeded in finding the operating point.

tr_q=1.23839e-009 FROM 8.16876e-008 TO 8.2926e-008
tf_q=1.67466e-009 FROM 1.01861e-007 TO 1.03535e-007
```

Figure 16: P-latch layout – rise/fall measurement.

6.4 N-Latch

A screenshot of a SPICE Error Log window titled "SPICE Error Log: C:\Users\hp\OneDrive\IC\assignment 2\p_latch.log". The log contains the following text:

```
Circuit: *** SPICE deck for cell p_latch(sch) from library p_LATCH

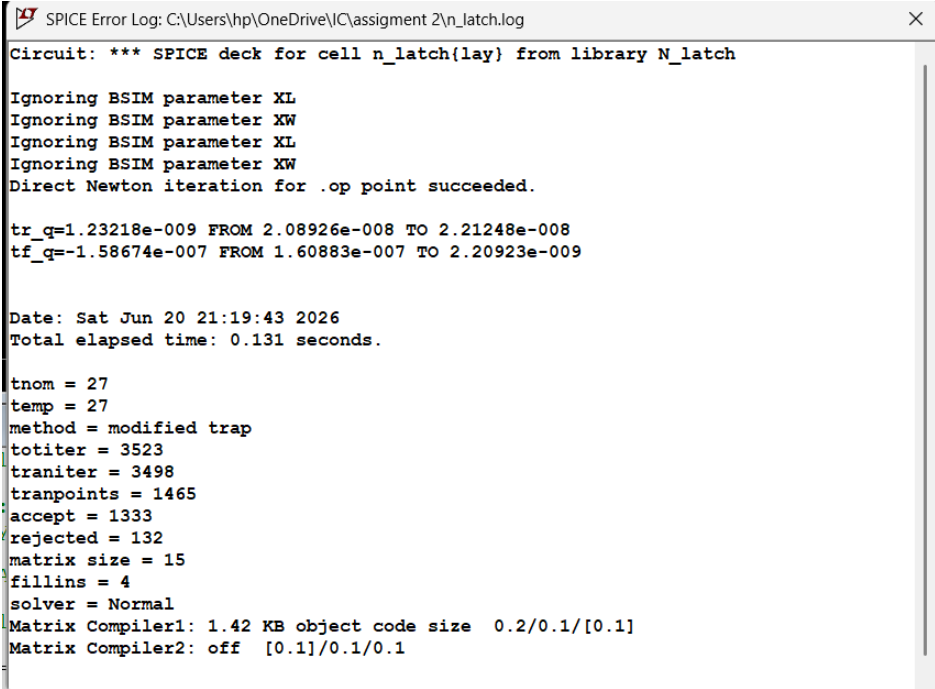
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Direct Newton iteration for .op point succeeded.

tr_q=1.19693e-009 FROM 8.16283e-008 TO 8.28252e-008
tf_q=1.65454e-009 FROM 1.01781e-007 TO 1.03435e-007

Date: Sat Jun 20 20:59:52 2026
Total elapsed time: 0.186 seconds.

tnom = 27
temp = 27
method = modified trap
totiter = 3693
traniter = 3647
tranpoints = 1410
```

Figure 17: N-latch schematic – rise/fall measurement.

A screenshot of a SPICE Error Log window titled "SPICE Error Log: C:\Users\hp\OneDrive\IC\assignment 2\n_latch.log". The log contains the following text:

```
Circuit: *** SPICE deck for cell n_latch(lay) from library N_latch

Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Direct Newton iteration for .op point succeeded.

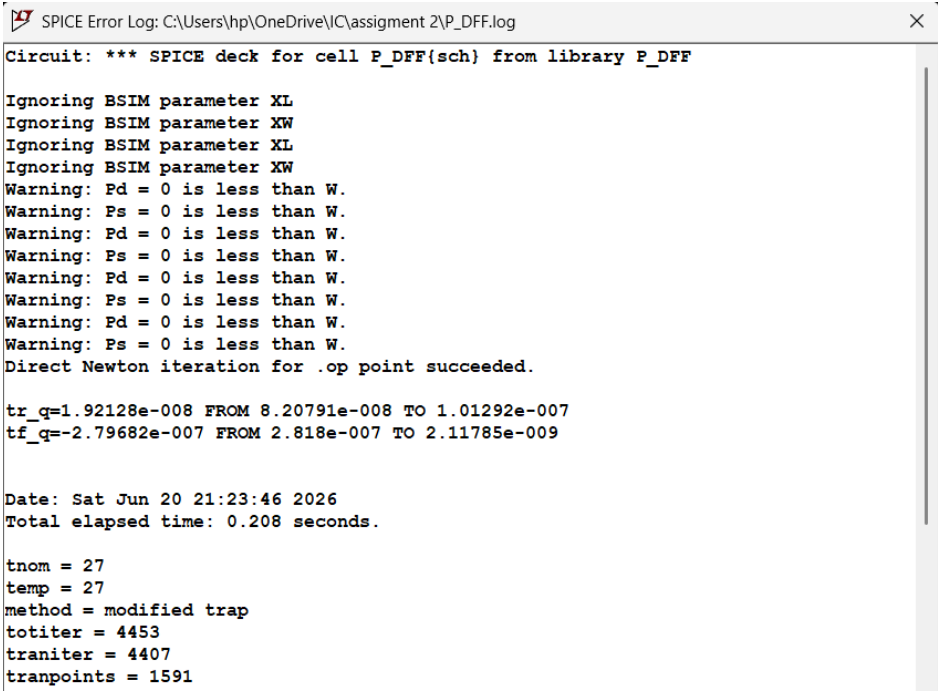
tr_q=1.23218e-009 FROM 2.08926e-008 TO 2.21248e-008
tf_q=-1.58674e-007 FROM 1.60883e-007 TO 2.20923e-009

Date: Sat Jun 20 21:19:43 2026
Total elapsed time: 0.131 seconds.

tnom = 27
temp = 27
method = modified trap
totiter = 3523
traniter = 3498
tranpoints = 1465
accept = 1333
rejected = 132
matrix size = 15
fillins = 4
solver = Normal
Matrix Compiler1: 1.42 KB object code size 0.2/0.1/[0.1]
Matrix Compiler2: off [0.1]/0.1/0.1
```

Figure 18: N-latch layout – rise/fall measurement.

6.5 D Flip-Flop



```
SPICE Error Log: C:\Users\hp\OneDrive\IC\assignment 2\P_DFF.log
Circuit: *** SPICE deck for cell P_DFF{sch} from library P_DFF

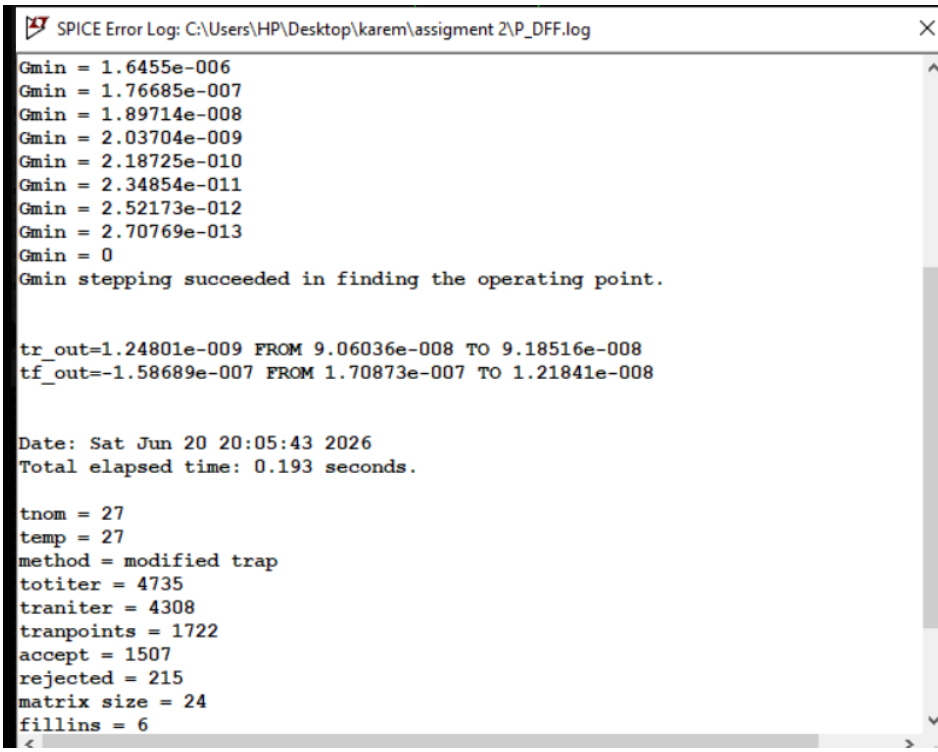
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Ignoring BSIM parameter XL
Ignoring BSIM parameter XW
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Warning: Pd = 0 is less than W.
Warning: Ps = 0 is less than W.
Direct Newton iteration for .op point succeeded.

tr_q=1.92128e-008 FROM 8.20791e-008 TO 1.01292e-007
tf_q=-2.79682e-007 FROM 2.818e-007 TO 2.11785e-009

Date: Sat Jun 20 21:23:46 2026
Total elapsed time: 0.208 seconds.

tnom = 27
temp = 27
method = modified trap
totiter = 4453
traniter = 4407
tranpoints = 1591
```

Figure 19: D Flip-Flop schematic – rise/fall measurement.



```
SPICE Error Log: C:\Users\HP\Desktop\karem\assignment 2\P_DFF.log
Gmin = 1.6455e-006
Gmin = 1.76685e-007
Gmin = 1.89714e-008
Gmin = 2.03704e-009
Gmin = 2.18725e-010
Gmin = 2.34854e-011
Gmin = 2.52173e-012
Gmin = 2.70769e-013
Gmin = 0
Gmin stepping succeeded in finding the operating point.

tr_out=1.24801e-009 FROM 9.06036e-008 TO 9.18516e-008
tf_out=-1.58689e-007 FROM 1.70873e-007 TO 1.21841e-008

Date: Sat Jun 20 20:05:43 2026
Total elapsed time: 0.193 seconds.

tnom = 27
temp = 27
method = modified trap
totiter = 4735
traniter = 4308
tranpoints = 1722
accept = 1507
rejected = 215
matrix size = 24
fillins = 6
```

Figure 20: D Flip-Flop layout – rise/fall measurement.

6.6 Schematic vs. Layout – Summary

Table 1: Rise time and fall time: schematic vs. layout (ns).

Block	t_r Sch.	t_f Sch.	t_r Lay.	t_f Lay.
Pass Gate	1.295	-158.968	1.323	-158.954
Tri-State Inverter	-159.367	1.223	-159.328	1.284
P-Latch	1.197	1.655	1.238	1.675
N-Latch	1.197	1.655	1.232	-158.674
D Flip-Flop	19.213	-279.682	1.248	-158.689

Layout parasitics slightly increase rise time across most blocks, consistent with the added interconnect and diffusion capacitance from the physical layout. The negative entries above (Pass Gate t_f , Tri-State Inverter t_r , N-Latch layout t_f , and D Flip-Flop t_f) come directly from the LTspice `.measure` log: in each of these cases the target threshold crossing was detected before the trigger crossing in simulated time, which is not physically meaningful and points to a measurement-setup artifact rather than a real transition time.

7 Conclusion

The pass gate and tri-state inverter combine to form a transparent latch; clocking two such latches (P-latch, N-latch) in opposite phase and cascading them produces a master-slave D Flip-Flop that updates Q once per clock cycle. Layout-extracted simulations show the expected small increase in rise time relative to the schematic due to interconnect parasitics.